

1 **Abstract**

2 Urban green spaces are critical for maintaining biodiversity and supporting wildlife, especially in
3 heavily urbanized environments. In temperate regions, winter brings significant challenges for
4 resident bird species, which can be compounded or alleviated in urban regions. To better
5 understand how birds use urban habitats during winter, we experimentally translocated Northern
6 Cardinals (*Cardinalis cardinalis*) among sites spanning an urbanization gradient in Montreal,
7 Canada, and used VHF telemetry to quantify their home range size, habitat selection across
8 spatial scales and habitat associations. Cardinal populations increased across the northeastern
9 United States and southeastern Canada in the past 100 years with most cardinals restricted to
10 cities, likely due to the presence of bird feeders. Cardinals in the most urbanized study site had
11 larger home ranges compared to birds wintering in the suburban and peri-urban sites. Habitat
12 selection occurred at the smallest scales measurable (25-50 m) across all sites
13 (urban/suburban/peri-urban) implying very fine-scale selection. This emphasises the importance
14 of microhabitat features such as vegetation structure and bird feeder presence. Habitat
15 associations varied across the urban gradient, except for proximity to bird feeders which was
16 consistently related to habitat use by cardinals. In less urbanized areas, birds selected for
17 vertically complex vegetation, likely because cardinals select shrubby, edge habitat. In more
18 urbanized areas, birds were more likely to use areas with a greater density of roads, implying
19 cardinals select the most residential areas. Cardinals exhibited behavioral plasticity in response
20 to urbanization, suggesting that fine-scale ecological understanding can help guide locally
21 tailored green space management aimed at supporting urban-adapted species year-round.

Keywords: behavioral plasticity, fine-scale movement, habitat association, home range size, manual VHF telemetry, non-breeding period, scale of habitat selection, urbanization gradient

Lay summary

- Urban green spaces can help wildlife survive in cities, but little is known about how birds use these spaces during the winter months in northern climates.
- We tracked Northern Cardinals in Montreal, Canada, using small radio transmitters to follow where they move during winter.
- By tracking these birds, we quantified how much space they used and what kinds of places they favoured in areas ranging from greener neighbourhoods to denser parts of the city.
- Cardinals spending the winter in the most built-up areas used larger territories and relied on resources found close by, such as vegetation features and human-provided food sources like bird feeders.
- Overall, cardinals showed a strong ability to adjust to city life, suggesting that well-designed urban green spaces can support urban-adapted birds year-round.

Introduction

Urbanization is accelerating worldwide (Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022), altering landscapes and bringing novel challenges to wildlife including habitat fragmentation, habitat loss, and chronic pollution (Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023). Future global urban expansion could lead to an additional 11-33 million hectares of habitat loss by 2100, with a disproportionately large increase in habitat fragmentation (Li et al. 2022).

This habitat loss and fragmentation can in turn reshape wildlife spatial ecology, including patterns of home range size and movement. A meta-analysis showed both birds and mammals had smaller home ranges in urban areas and that urbanization restricts spatial movements for most animals (O'Donnell and delBarco-Trillo 2020). During the breeding period, the communication distance of *Zonotrichia leucophrys* (White-crowned Sparrows) in the San-Francisco Bay was highly predictive of its territory size, highlighting that in the urban landscape, noise pollution may drive songbirds into smaller home ranges in the breeding season (Simberloff et al. 2023). Nonetheless, most research on urban birds occurs in the breeding season, and there is far less in winter, and those studies that do exist contrast with breeding season studies. This knowledge gap is important because birds often exhibit different behaviours outside of the breeding season. For example, *Cardinalis cardinalis* (Northern Cardinals) form winter flocks (while vigorously defending their territories against conspecifics during the breeding season) and alter their patterns of space use during the non-breeding season in Kentucky (Ritchison and Omer 1990). Consequently, factors influencing habitat use during winter may differ substantially from those operating during the breeding season.

Birds respond to these urban pressures through behavioural adjustments, reflecting both challenges and opportunities, including increased predation pressure, heightened disturbances, novel anthropogenic resources and human-made structures. To compensate for disturbances from noise, light and human activity, some birds may change their nesting behaviour—for example by nesting higher in the canopy (e.g., *Pica pica* (Eurasian Magpie); Šálek et al. 2020) or using off-ground artificial surfaces increasing fitness (e.g., *Junco hyemalis* (Dark-eyed Junco); Bressler et al., 2020). Urban habitats can also provide foraging opportunities which can be exploited by some species like *Falco peregrinus* (Peregrine Falcon). *F. peregrinus* was shown to use buildings

as perches and nesting sites, capitalizing on disoriented migratory birds wandering near lit glass structures (Kauffman et al. 2003; DeCandido and Allen 2006; Altwegg et al. 2014). Although some species can adapt and thrive in urban areas, urban tolerance varies across species and across annual cycles, with urban tolerance peaking during the winter months (potentially due to the urban heat island effect) (Callaghan et al. 2021).

Habitat use in the city can differ significantly from that in rural environments, even within a single species. These differences can reflect hierarchical habitat selection (Johnson 1980), as animals select habitat at multiple spatial scales—from geographic range (first order), to the home range of an individual (second order), to the selection of specific patches for daily activities (third and fourth orders). In the urban context those decisions are influenced by novel pressures such as human disturbances, artificial structures, and altered resource availability. Quantifying habitat use by comparing used locations with what is available in the habitat can provide valuable insights into how wildlife responds to urbanization and identify habitat features that are more strongly associated with both occurrence and space use (Manly et al. 2002; Poirier et al. 2024).

Urban birds may outperform rural birds during winter, when natural resources are less available and weather is harsher. Cities can provide a more stable wintering environment for bird species that use feeders or that are generalists or omnivores (i.e., corvids), which exploit anthropogenic resources in winter (Jokimäki et al. 2022). In other cases, cities can serve as refuge during periods of extreme weather, as local colonization of cities by birds increased during the colder conditions due to cities acting as heat islands (Latimer and Zuckerberg 2021). Nonetheless, occupancy dynamics in temperate regions during winter are highly species-specific, with thermal tolerance playing an important role in determining which species would be able to persist in

urban environments during this season. Moreover, energy demands for small resident passerines increase with latitude in winter, meaning that birds spending their winter in regions such as southern Quebec must consume significantly more energy to maintain their body temperatures than those wintering farther south—which is especially challenging given the scarcity of resources during the cold months (Fristoe 2015). Despite these physiological and environmental challenges, along with the potential added effects of urbanization, winter remains an underrepresented season in experimental, ecological research (Dinh et al. 2023). This gap in knowledge highlights the importance of investigating how birds are occupying urban habitats in winter, particularly in temperate regions where winter survival is a major limiting factor for resident bird species.

To address this gap, we studied *Cardinalis cardinalis* (Northern Cardinal) across an urban gradient in one of Canada's largest cities, Montreal, Quebec. Cardinals are large songbirds (weighing 40 to 45 grams), which allows for deployment of radio tags capable of lasting almost the entirety of the non-breeding season. In the last 100 years, cardinals expanded their distribution range northwards, and have been anecdotally closely associated to urban and suburban landscapes in those areas (Beddall 1963; Smith et al. 2011; Robert et al. 2019). They are able to thrive and persist across a wide range of urbanization gradients, from suburban neighbourhoods to densely populated areas of the city. Cardinals are now resident in southern Quebec, a region of extreme cold (sometimes reaching -27 °C with wind chill; Environment and Climate Change Canada) and limited natural food sources during the winter months (Government of Canada 2004; Turcotte and Desrochers 2005). As part of a translocation experiment, we captured and released individuals at sites distributed along this urbanization gradient in Montreal to evaluate how they are using such habitat. We tracked weekly

movements to assess how the birds adjust their movements and habitat use after being displaced into environments with varying levels of urbanization.

This study has four objectives: (i) investigate how cardinals' home range size varies across an urbanization gradient through the winter season, (ii) quantify the scale at which cardinals are associated with certain factors of their habitat, (iii) determine the relative strength of anthropogenic and natural vegetation to predict habitat use by cardinals during the non-breeding season, and (iv) document the spring movements from non-breeding to breeding grounds. We predicted that wintering cardinals require larger home ranges (both core and total) and select habitats at larger scales when in dense urban locations due to food sources being more scattered (Havlíček et al. 2022). In addition, we predicted that locations with bird feeders, lower population density and a complex vertical vegetation structure would be associated with a higher probability of cardinals. Finally, we predicted that cardinals would transition from their breeding sites in forested/canopy covered areas towards residential areas in the fall to take advantage of bird feeders during winter, reversing the movement in spring.

Materials and methods

Study locations

The island of Montreal is located in southern Quebec, Canada, with a population of ~2 million people coupled with high population density. The city's climate is comprised of four distinct seasons with cold winters (temperatures can range between -23 °C and 4 °C during the month of January) and warm summers (with temperatures sometimes exceeding 30 °C; WeatherSpark). This research was conducted at three study sites (Figure 1) with varying levels of urbanization. Each site includes two release locations randomly assigned to the individuals. The peri-urban site is located on the western end of the island of Montreal at the McGill Bird Observatory (MBO;

45°25'51.4"N 73°56'18.5"W). The MBO is part of the 30 km² *Grand parc de l'Ouest*. The suburban site is located in the Town of Baie d'Urfé (45°24'39.2"N 73°55'50.0"W). The urban study site is located in the Notre-Dame-de-Grâce neighbourhood of the city (45°27'35.7"N 73°38'30.2"W). Because cardinals can be found in all three sites, we assumed that the translocated birds would be able to survive and thrive at each location.



Figure 1. Study locations on the Island of Montreal A) The peri-urban site is at the McGill Bird Observatory (2% of the area covered by streets and/or roads). B) The suburban site (5% of the area covered by streets and/or roads) is located in the residential Town of Baie d'Urfé. C) The urban site (13% of the area covered by streets and/or roads) is located in Notre-Dame-de-Grâce neighbourhood.

The three translocation sites differed significantly in their level of urbanization, following a Kruskal-Wallis test and a Bonferroni-corrected pairwise Wilcoxon test ($p < 0.001$) based on population density data from the Cencensus Program R package (von Bergmann et al. 2021), using census data from the Government of Canada for the year 2021. All three sites are characterized by relatively high levels of green space compared to the broader urban context of Montreal. The gradient used here ranges from the higher percentages of canopy covers in the peri-urban site to the median canopy cover for the Montreal area (~ 26%) (HealthyDesign.City 2021). As such, this gradient does not fully capture the extent of urbanization present across the entire city and should be interpreted as a localized urbanization gradient. We did try translocating cardinals to an area with lower canopy cover early on, but due to two immediate mortalities we decided to omit this site.

Radio tagging, banding & translocation

All birds were caught during routine fall migration monitoring at MBO and were fitted with colour bands and Motus NanoTags following standard banding and morphological measurements (see Supplemental Information). Cardinals translocated to sites away from the peri-urban location (suburban and urban), as well as control birds held at the peri-urban site, underwent a holding time of 45 minutes before release to standardize treatment amongst individuals. For the winter of the year 2022-2023 (winter 1), we were able to consistently track 20 birds released as control at the peri-urban site, which served as a pilot winter to assess the feasibility of the project. During the winter of 2023-2024 (winter 2), we tagged and banded 45 birds, with 15 individuals translocated to the urban site, 15 translocated to the suburban site and 15 released as controls at the peri-urban site. In 2024-2025 (winter 3), we tagged and banded 42 birds, with 5 individuals released as controls at the peri-urban site, 16 individuals translocated to the suburban

site and 21 individuals translocated to the urban site. During the reproductive season, cardinals with tags (weighing 1.6 g [double the 0.8 g of the tags used in the present study]) attached defended their nests less vigorously, but did not change their provisioning effects (Barron et al. 2013). Cardinal nests with tagged parents also had lower daily survival rates and were less likely to fledge offspring (Barron et al. 2013). However, these tag effects were relatively modest and observed across diverse habitats, making it improbable that they would only manifest in our dense urban site while remaining undetected in peri-urban and suburban locations. Birds were banded under Animal Use Protocol 2007-5447 from McGill University and federal banding permits 10973, 10973B, 10973D issued by the Canadian Wildlife Service, and radio-tagged under Animal Use Protocol 22BF01 and 23BF01 from Environment and Climate Change Canada.

Cardinals' abundance and seasonal distribution

We used the number of cardinals recorded per party hour in Audubon Christmas Bird Count data to compare the historical abundance of cardinals in Quebec, Canada, against the states of Maine, New York, New Hampshire and Vermont (United States) (National Audubon Society 2020).

Non-breeding season tracking and home range estimations

The birds were monitored using an array of automated Motus towers and handheld telemetry receivers. We conducted manual radio tracking of each individual two to three times per week between the months of October and May each year. The areas monitored were ~5 km² (500 ha) at the peri-urban site, ~6 km² (600 ha) for the suburban site and ~18 km² (1800 ha) for the urban site. Tracking effort and spatial coverage differed among sites due to logistical constraints associated with weather conditions and the structure of the landscape. At the peri-urban site, the absence of road networks and lower levels of signal interference allowed for the tracking team to

193 move more freely and locate the individuals more rapidly, resulting in a smaller monitored area.
194 In contrast, tracking the birds in suburban and urban areas meant that the tracking team's
195 movements were constrained by road access, built infrastructure, private land and higher levels
196 of radio interference. In addition, individuals released at the suburban and urban sites often
197 moved farther away from their release location, requiring the tracking team to search across
198 larger areas to maintain contact with tagged birds. These differences in spatial coverage reflect
199 practical limitations of tracking birds in the heterogeneous landscape of urban environments and
200 should be considered when comparing movement patterns across sites. The people tracking the
201 birds by hand (trackers) aimed to record at least one GPS location for each individual per week
202 using a Garmin ETREX 22x GPS. Trackers would typically spend 5 hours tracking per session,
203 twice per week, and started by first tracking down newly relocated birds and/or birds with known
204 recent locations. Once these individuals are found, trackers would move on to expand their
205 search areas to try and find missing individuals. Details on the tracking method used and
206 estimations of precision can be found in the Supplemental Information section.

207 For all three study sites, we calculated the average home range size using the Kernel Density
208 Estimation (KDE). The home range derived from the utilization distribution (UD) represents the
209 minimum area containing a specified proportion of an individual's space use (in the present case,
210 the cardinals' GPS location recorded in the field) with the kernelUD function from the
211 adehabitatHR: Home Range Estimation R package version 0.4.22 (Worton 1989; Calenge 2023,
212 2024). For example, the 95% home range represents the smallest area in which there is a 95%
213 probability of finding the individual. We selected the 50% and 95% contours to investigate
214 respectively the core area used within the home range of the birds and the broader home range
215 excluding the most extreme locations (outliers). A minimum of 5 GPS locations were required to

adequately run the function to calculate the KDEs meaning that individuals with fewer points than this threshold were excluded. Between 5 and 48 points per individual were used to estimate the KDEs, with the average number of points per bird being 18.

Scale-of-effect

We first identified the scale-of-effect (i.e., the spatial extent at which landscape variables have the strongest relationship with a species' response') by extracting variables at four spatial scales (25, 50, 100, 200 meters; Martin & Fahrig, 2012). These spatial scales were chosen because they cover a wide range of possible scale-of-effect that are relevant in urban ecology, considering that habitat heterogeneity is scale-dependent. However, because scale-of-effect analyses identify the spatial extent that best predicts responses to the tested range, these values should be interpreted as indicating generally fine or coarse spatial responses rather than precise ecological thresholds. For each scale we ran generalized linear mixed models (GLMMs) at each study location using all five predictor variables as well as year, site, and individual ID as random effects. We ranked models using Akaike Information Criterion (AICc) values, which take into account small sample sizes (Burnham and Anderson 2002). The best model was identified as having the lowest AICc value. We also calculated the $\Delta AICc$ to evaluate competing models following Burnham and Anderson (2002) (models with a $\Delta AICc < 2$ were considered to have substantial support, competing with the best model and models with a $\Delta AICc > 10$ were deemed to have no support, not competing with the best model). We acknowledge that low per-individual sample sizes might reduce the precision of the scale-of-effect estimates, as models fitted on few observations per individual could be susceptible individual-level variation driving apparent scale preferences. To try mitigating this, we included the individuals' IDs as random effect, and we interpret scale-of-effect results at the population level rather than an individual level habitat selection.

Because the four buffer sizes represent nested spatial extents, we expect the variables extracted from these buffers to be positively correlated. We assessed cross-scale correlations for each predictor type using Pearson's "r". Correlations were highest for population density ($r = 0.96$ - 0.99 across all sites), moderate for mean vegetation heights ($r = 0.37$ - 0.94) and road length ($r = 0.32$ - 0.95), and lowest for SD vegetation height ($r = 0.33$ - 0.84), decreasing with increasing scale separation in all cases. Variables from different scales were never included within the same model—each candidate model uses a single buffer size—so cross-scale correlations do not inflate within-model multicollinearity (all VIF < 10). The buffer comparison therefore constitutes a scale-selection exercise rather than a multi-scale additive approach. Once the scale-of-effect was determined for each location, we conducted a resource selection function.

Resource Selection Function

We compared 'used' vs. 'available' habitat using a resource selection function for cardinals during the non-breeding season and across an urban gradient. We used environmental and anthropogenic variables extracted from available spatial data as well that collected in the field. This included mean and standard deviation (SD) of vegetation height (1m resolution) from OpenForest (Government and Municipalities of Quebec 2016), distance to nearest bird feeder (collected through systematic field observations), population density data (census unit) using the Cancensus R package (von Bergmann et al. 2021) and spatial data on the streets and road from the island of Montreal (Ville de Montréal 2018). Unused locations were derived at a 1:1 ratio by creating random locations within 300 meters of the used location, at a random angle. We used the matched case-control design elaborated by Fortin and have applied a 300 meters buffer to reflect the median daily bird movement (Fortin et al. 2005, 2009). Because this is a population-level resource selection function, in which individual IDs are included as random effects and used-

262 available pairs from all individuals are evaluated together, there is no required minimum number
 263 of locations per individual, as it would be the case for an individual-level RSF. The average
 264 number of true observations for each individual ranges from 5 to 48 with an average number of
 265 points of 18. Having a ratio of 1:1, the same numbers apply for the random observations. The
 266 data extracted was scaled and centered before running the models. We built our candidate model
 267 set to compare the relative strengths of anthropogenic and natural vegetation variables to predict
 268 cardinal habitat use. This included four *a priori* hypothesis-based models including, (1) the “Full
 269 model” (Used ~ Mean vegetation height + Sd vegetation height + Population density + Road
 270 length + Distance to nearest feeder + (1|Winter) + (1|Site) + (1|ID)), including all of the
 271 variables, plus the winter (years of sampling), relocation site and bird ID as random effects to
 272 account for the additive importance of anthropogenic and natural vegetation variables, (2) the
 273 “Vegetation height” model (Used ~ Mean vegetation height + (1|Winter) + (1|Site) + (1|ID)),
 274 assuming the vertical height of vegetation was the dominant variable related to habitat use by
 275 cardinals, (3) the “Sd vegetation height” model (Used ~ Sd vegetation height + (1|Winter) +
 276 (1|Site) + (1|ID)), assuming that cardinal habitat use is related to vertical vegetation
 277 composition, and (4) the “Anthropogenic” model (Used ~ Population density + Road length +
 278 Distance to nearest feeders + (1|Winter) + (1|Site) + (1|ID)), which assumed that cardinal habitat
 279 use would be related to anthropogenic variable. We also included a “Null” model, with random
 280 effects only (Used ~ + (1|Winter) + (1|Site) + (1|ID)). We ranked all five models according to
 281 the corrected Akaike Information Criterion (AICc) following the method previously described by
 282 Burnham and Anderson (2002). We used Variance Inflation Factor (VIF) to test for
 283 multicollinearity (variables with VIF values > 10 were considered to have problematic
 284 collinearity and were removed) among predictor variables and Hosmer-Lemeshow as a

goodness-of-fit test (Hosmer and Lemeshow 1980). VIFs ranged between 1.05 and 3.31 across all predictor variables. All statistical analysis were performed using Program R version 4.4.2 (R Core Team 2024).

Results

Historical and seasonal context of Cardinals in southern Quebec

To place our winter telemetry results in the context of the species' recent expansion into southern Quebec and its seasonal dynamics within the study region, we summarized long-term regional abundance and local banding records. In southern Quebec, cardinals have increased in abundance since the 1970s, reflecting the species' well-documented northward range expansion from the northeastern United States into southern Quebec beginning in the 1960s (Figure 2).

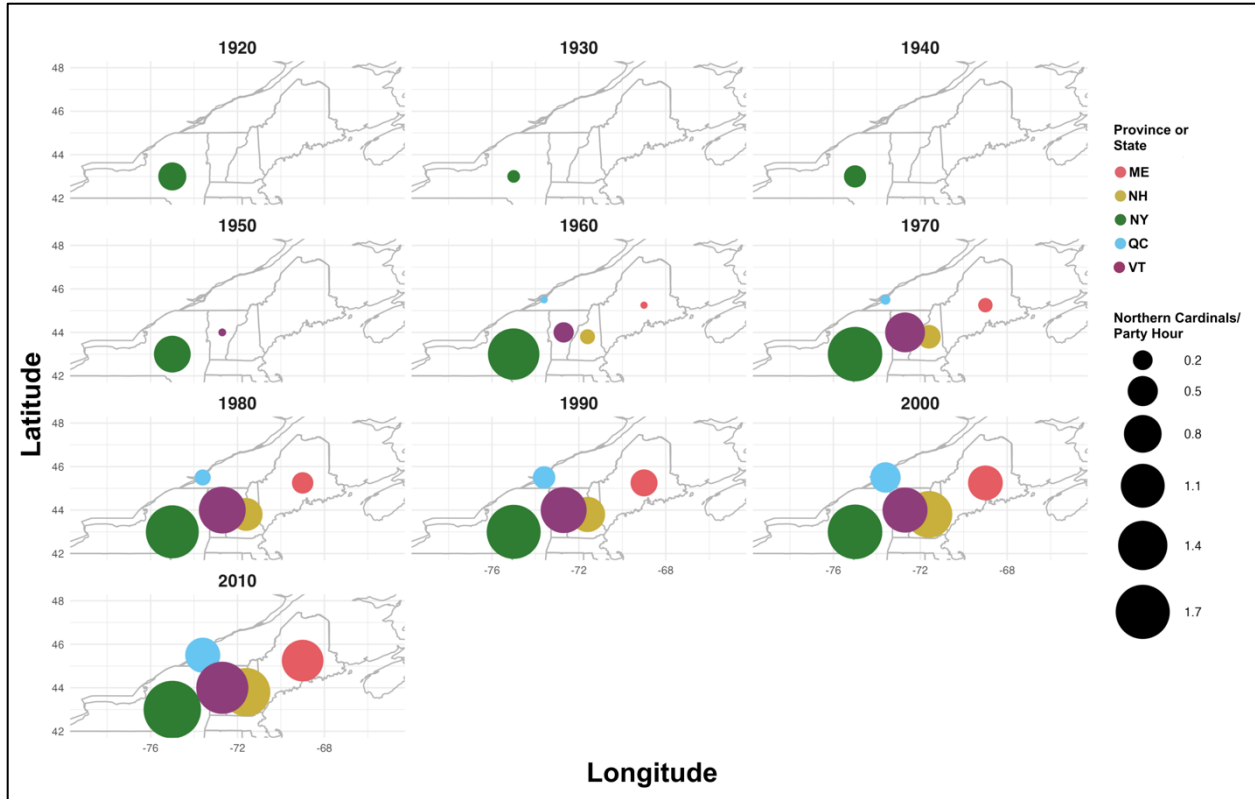


Figure 2. Number of cardinals observations per party hour between the 1920s and the 2010s according to state (United States) or province (Canada) as part of the Audubon Christmas Bird Count.

Nineteen years of banding data from the McGill Bird Observatory also revealed distinct seasonal patterns in captures (Figure 3). Capture rates were higher during the fall monitoring program (~0.75 birds/h) than during the spring program (~0.27 birds/h), accompanied by a greater proportion of unbanded individuals, indicating an influx of new birds in late fall. Although most fall captures were juveniles or individuals of unknown age (Figure S1B, C), adults also exhibited peaks in mid-May and late July (Figure S1A), suggesting that both post-fledging dispersal and local movements between breeding and wintering areas contribute to seasonal variation in captures.

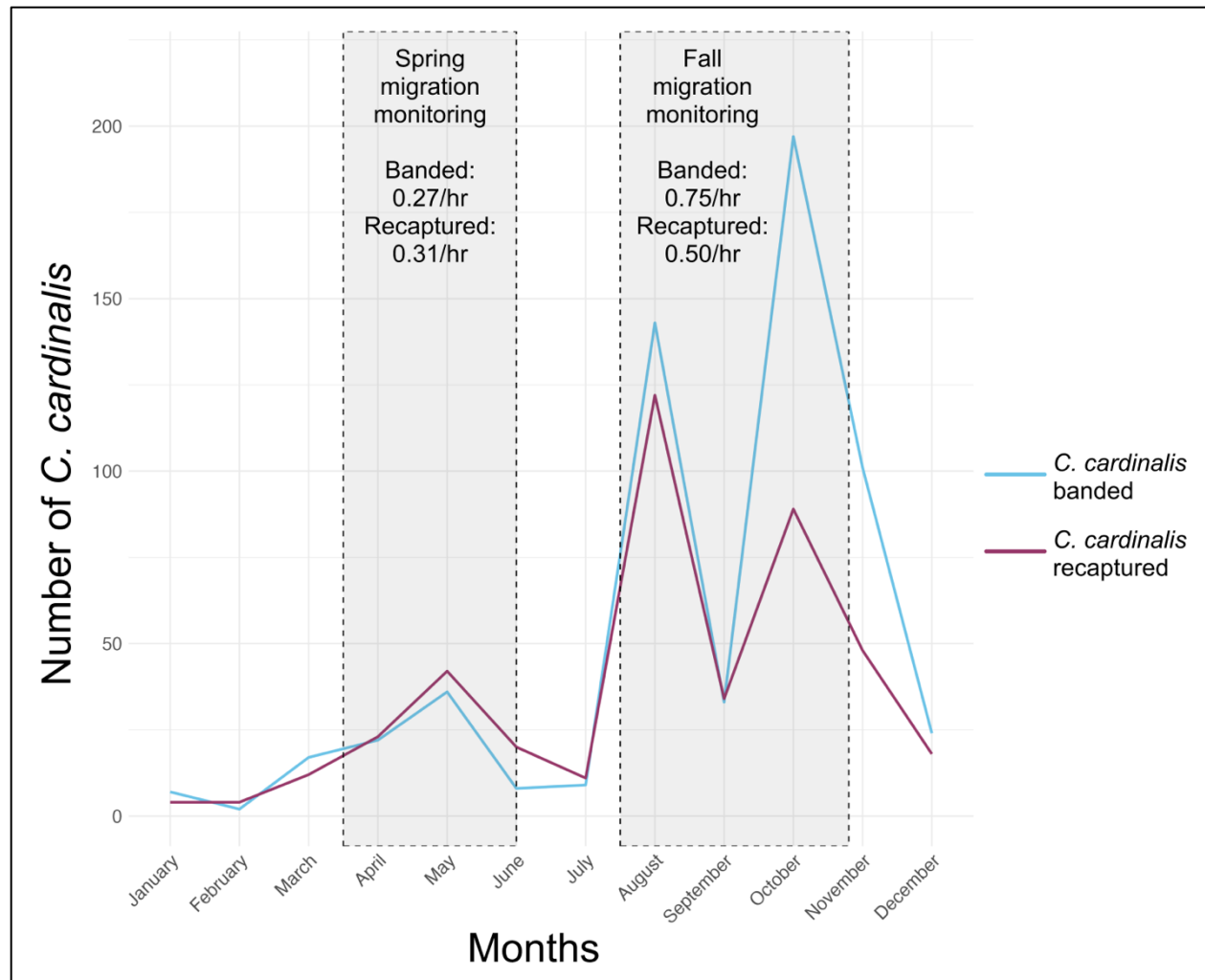


Figure 3. Number of cardinals banded and recaptured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada. The grey boxes highlight the duration of the spring and fall migration monitoring programs, along with the approximate number of cardinals banded and recaptured per hours of mist-net being opened during the season.

Telemetry tracking of cardinals across an urban gradient

Across three consecutive winters, we were able to track multiple individuals at all three sites and gather valuable information regarding cardinals' survival through the study (Table 1).

Table 1. Number of cardinals tracked 1 and 3 months after initial release, with apparent survival (48.15% average) and fated survival (including individuals presumed alive but outside the study area; 71% average), including known mortality events/probable causes across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of cardinals	# tracked 1 month (apparent survival)	# tracked 3 months (apparent survival)	Missing individuals after 3 months	Confirmed mortality events and probable cause of death
Peri-urban	41	34 (83%)	28 (68%)	13	6 Predation (n = 4) Unknown (n = 2)
Suburban	31	19 (61%)	18 (58%)	13	4 Predation (n = 2) Disease (n = 1) Unknown (n = 1)
Urban	36	11 (31%)	6 (17%)	30	3 Predation (n = 1) Window collision (n = 2)

Telemetry tracking was conducted 2 to 3 days per week by three people between October and May 2022-2024, for a total of ~288 days of tracking per winter. We recorded a total of 688 unique detections at the peri-urban site, 411 at the suburban site, and 265 at the urban site. The tracking team covered the maximum amount of ground possible during each outing, to record as many individuals as feasible.

Core areas and home ranges of cardinals across an urban gradient

We estimated core areas and home ranges for individuals with ≥ 5 recorded GPS locations (35 individuals from the peri-urban site, 13 from the suburban site and 9 from the urban site; Figure 4). Among the 58 individuals with ≥ 5 recorded GPS locations, core and total home range sizes were not normally distributed (Shapiro-Wilk tests: core, $W = 0.624$, $p < 0.001$; total, $W = 0.698$, $p < 0.001$), so areas were transformed using a logarithm prior to analysis (Figure 4; Table 2).

Table 2. Number of cardinals that could be included in the analyses along with their respective core areas and home ranges including standard errors across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of cardinals included in the analyses	Core area (km ²)	SE of the core area (km ²)	Home range (km ²)	SE of the home range (km ²)
Peri-urban	36	0.15	0.04	0.61	0.13
Suburban	13	0.08	0.02	0.56	0.11
Urban	9	0.28	0.11	1.53	0.52

After transformation, the data met assumptions of normality and variances were homogeneous. There was no significant difference in core home range size among sites, although the effect approached significance (ANOVA: $F_{2, 55} = 2.69$, $p = 0.077$). Post-hoc comparisons show a trend towards the cardinals at our urban site to have larger core home range sizes compared to the suburban and peri-urban sites (Figure 4). There was a significant difference among the three study sites regarding total home ranges (ANOVA: $F_{2, 55} = 3.45$, $p = 0.04$). Post-hoc TukeyHSD comparisons showed that the only significant difference was between the total home ranges of cardinals at our urban site being larger than cardinals at the peri-urban site (mean difference = 1.24, 95% CI [0.09, 2.38], $p = 0.03$).

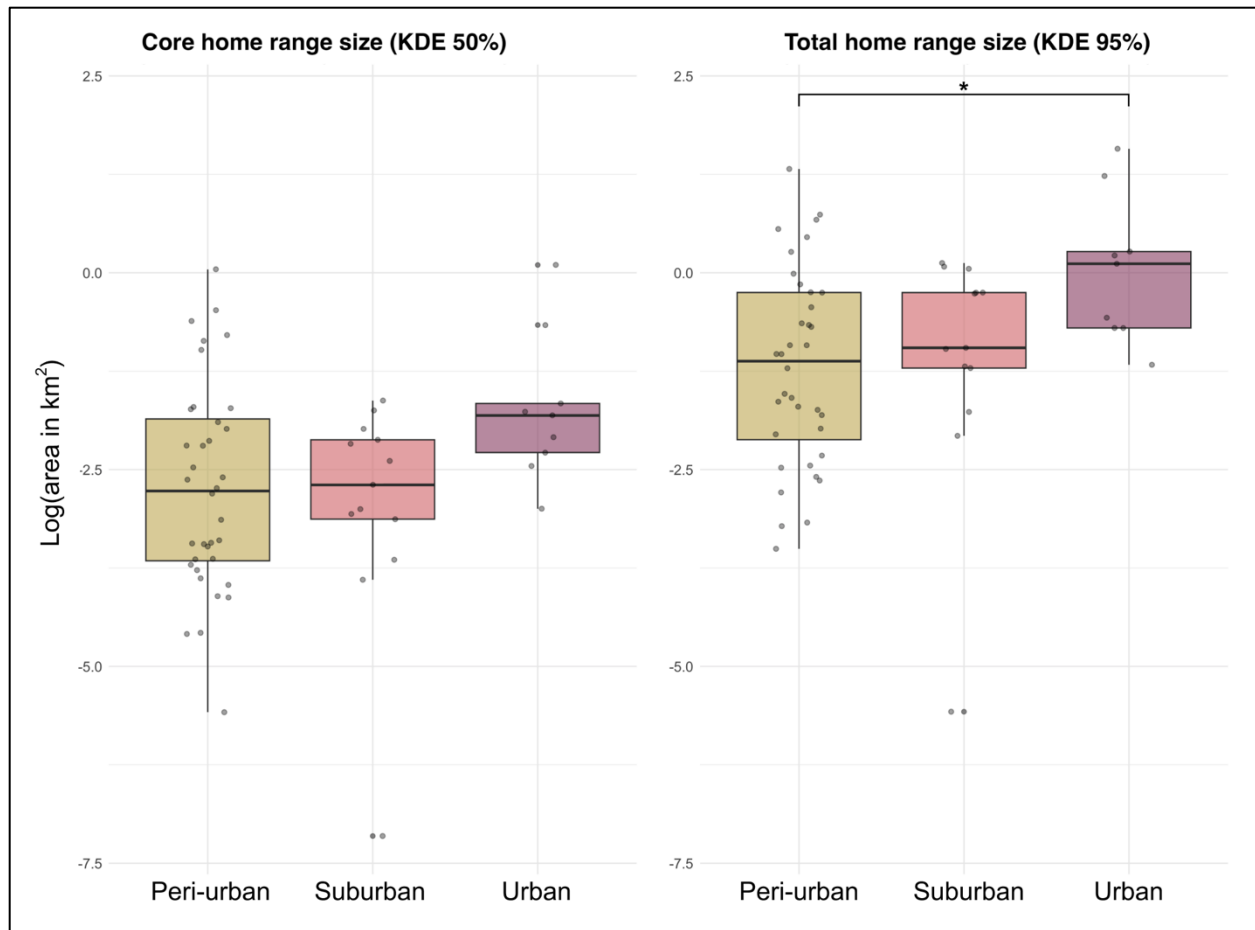


Figure 4. Boxplot comparison of the core (KDE 50%) and total (KDE 95%) home ranges of cardinals in three study sites (peri-urban, suburban and urban) in Montreal, Quebec, Canada during winters between the years 2022 and 2025. The area in km^2 was log transformed, and each data point corresponds to an individual's home range (area log transformed).

Scale of association to habitat

We compared AICc among models fit with landscape covariates measured at four different spatial scales (25, 50, 100 and 200 m radius around resighting locations). At the peri-urban site, the 50 m scale model best explained cardinal's movements ($\Delta \text{AICc} > 4$ relative to all other scales). At both the suburban and urban sites, the 25 m scale model performed best ($\Delta \text{AICc} > 10$ relative to all other scales). Because the landscape variables we measured at different spatial

scales are inherently correlated, these results indicate the scale at which habitat structure shows the strongest statistical association with cardinal's movements, rather than reflecting a direct causal scale of effect.

Resource selection: used vs. available habitats

Due to differences of urbanization level and scales of effect, resource selection function models were run separately for each study site. For the peri-urban site, the global model explained the most variation seen in the data ($> 10 \Delta \text{AICc}$ values between all other models, Supplementary Material Table S1). The full model reduced deviance by $\sim 11.3\%$ relative to the null model. For the suburban site, the top model was the anthropogenic covariates model (human population and road density), with the next best model (full model) $< 2\text{AICc}$ units from the top model (Supplementary Material Table S1). The anthropogenic covariates model reduced deviance by $\sim 16.6\%$ relative to the null model. Lastly for the urban site, the full model was the top model, but again the next best (anthropogenic covariates) was $< 1 \text{AICc}$ from the top model (Supplementary Material Table S1). The full model reduced deviance by $\sim 10.5\%$ relative to the null model. The best model for each study site was then used to generate prediction plots on the probability of presence of cardinals according to the model variables (Table 3).

Table 3. Model parameters for the three resource selection models predicting the presence/absence of 56 cardinals within (25 or 50 m-radius) areas based on the average vegetation height (in meters) and its standard deviation, along with human-related environmental features such as population density (in people/km²), road length (in meters) and distance to the nearest bird feeder (in meters). The study site, year (winter) of sampling and the individual's identification (ID) are included as random effects in each model.

Study Site	Variable	Coefficient estimate	Standard error	z	p
Peri-Urban	<i>Global model – 50 meters</i>				
	Mean vegetation height (m)	-1.272	0.153	-8.329	<0.001
	Standard deviation of the mean vegetation height	1.107	0.114	9.709	<0.001
	Mean population density (person/km ²)	-0.095	0.111	-0.858	0.391
	Road length (m)	0.023	0.110	0.207	0.836
Suburban	Distance to the nearest bird feeder (m)	-0.347	0.068	-5.067	<0.001
	<i>Anthropogenic covariates model – 25 meters</i>				
	Mean population density (person/km ²)	-0.486	0.135	-3.603	<0.001
	Road length (m)	1.238	0.136	9.138	<0.001
Urban	Distance to the nearest bird feeder (m)	-0.427	0.128	-3.342	<0.001
	<i>Global model – 25 meters</i>				
	Mean vegetation height (m)	0.252	0.180	1.399	0.162
	Standard deviation of the mean vegetation height	-0.458	0.191	-2.405	0.016
	Mean population density (person/km ²)	-0.212	0.131	-1.620	0.105
	Road length (m)	0.736	0.135	5.438	<0.001
	Distance to the nearest bird feeder (m)	-0.392	0.141	-2.772	0.006

The prediction plots including only the significant fixed effects for the resource selection functions of each locations reveal that the presence of cardinals is associated with different urban and habitat features according to the site they spend their winter in (Figure 5). In the peri-urban site, the probability of presence of the cardinals decreases as the mean vegetation height increases (Figure 5A), while the presence of the birds increases with greater variability in vegetation height (Figure 5B). The probability of presence of cardinals also seems to decrease as the distance to the nearest bird feeder increases (Figure 5C). In our suburban site, the probability of presence of cardinals shows little relationship with population density (Figure 5D), but increases as the mean road length increases (Figure 5E). Like our peri-urban site, the presence of the birds also decreases as the distance to the nearest feeder increases at the suburban site (Figure 5F). At the urban site, cardinals' probability of presence decreases with increasing vegetation height standard deviation, although uncertainty is high (Figure 5G). The presence of cardinals also increased with mean road length and decreased with increasing distance to the nearest feeders. Across sites, proximity to feeders and structural complexity of the infrastructures or the vegetation consistently influenced the probability of presence, yet specific relationships varied.

397 Taken together, these results indicate that winter habitat association by cardinals varies across the
398 urbanization gradient. While feeder availability appears to be consistently associated with a
399 higher probability of occurrence across all sites, the habitat features associated with cardinal
400 presence differed among the peri-urban, suburban and urban sites. This suggests that patterns of
401 habitat use are largely context dependent.

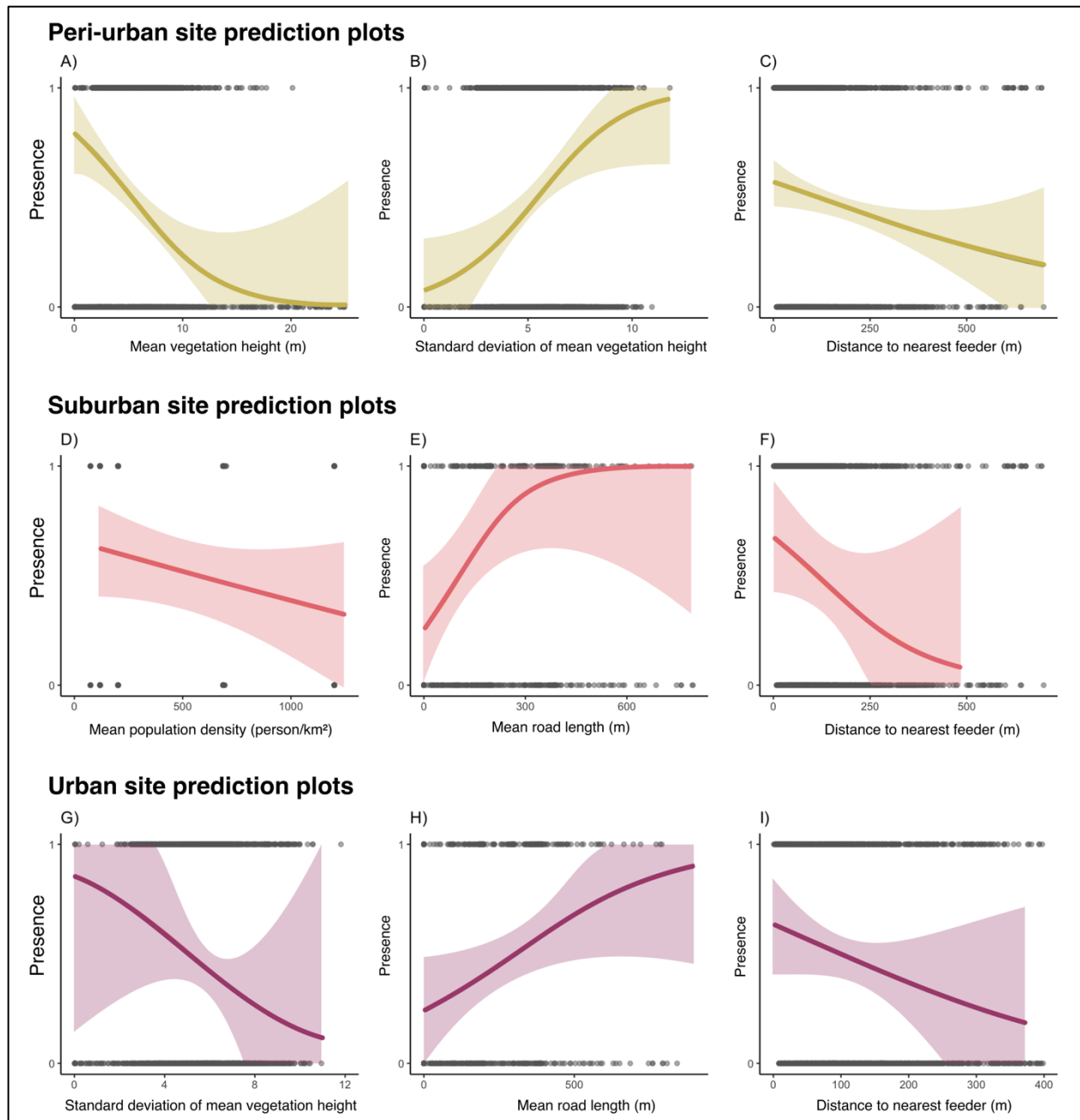


Figure 5. Probability of presence of cardinals in a given radius area, as predicted by a generalized linear mixed effects model. Shaded areas represent 67% confidence intervals around the predicted probabilities. The model's fixed effect variables are described in Table 2, and for each model, random effects were sampling years, translocation site and bird ID.

Discussion

This study examined winter space use, habitat associations and apparent survival of cardinals across an urban gradient. Although habitat associations differed among sites, broad patterns emerged. Individuals relocated at the urban site exhibited larger home ranges and lower apparent survival, than their counterparts at the suburban and peri-urban sites, while associations with environmental variables varied across the urbanization gradient. Proximity to bird feeders was consistently associated with cardinal occurrence across all sites, and individuals at the suburban and urban sites were primarily resighted in private UGS rather than larger public UGS. These findings suggest that winter habitat use by cardinals may vary according to local habitat conditions and resource availability, while anthropogenic food sources appear to be associated with space use across our range of urban environments. More broadly, our results indicate that private UGS may represent an important component of the habitat usage of the cardinals during the non-breeding season in southern Quebec.

Larger home ranges at urban site and the importance of bird feeders

Individuals at the urban site had the largest total (KDE 95%) home ranges (although we did not have a truly rural site, which may have also had a large home range). Because we couldn't capture the full spectrum of urbanization, the relationship to territory size may not be linear (i.e., larger home ranges in rural and urban areas compared to smaller home ranges in peri-urban and suburban areas). *P. domesticus* in urban landscapes occupied smaller home ranges compared to their rural counterparts through the breeding season (Vangestel et al. 2010). O'Donnell (2020) also suggested that birds in urban environments generally tend to have smaller home ranges. However, female cardinals venture out of their territory more often when they had supplemental feeding, compared to females that didn't have access to supplemental food, leading to a trend

towards larger home ranges (Humbird and Neudorf 2008). In our case, we found the opposite trend. By mapping the location of visible bird feeders, we obtained a coarse estimate of feeder density at all sites: the peri-urban site had ~13 feeders/km², the suburban site had ~35 feeders/km² and the urban site had ~12 feeders/km². Although these estimates provide useful context for interpreting site-level differences, they are not representative of all the feeders available and adequately maintained for the cardinals and should therefore be interpreted cautiously. One possible explanation is that the lower feeder density at the urban site, coupled with a potentially more challenging urban matrix, may have been associated with more extensive movements between reliable resource points. Because feeder density was not directly incorporated into our home range analyses, this interpretation should be considered as being on the speculative side. Conversely, the higher feeder density observed at the suburban site might have reduced the distance needed to move among reliable food resources. However, we want to point out that our study was not designed to test this mechanism directly. Future studies that are explicitly quantifying feeder availability and incorporating it into movement analyses would be valuable to evaluate whether variation in feeder density in urban areas is associated with winter home range size.

Lower apparent survival at urban site

While the apparent survival of cardinals was quite similar between the peri-urban site (on average 65% after 3 months) and the suburban site (on average 57% after 3 months), it was much lower at the urban location (on average 16% after 3 months). Our apparent survival rates (which may have been influenced by emigration, which we could not account for) for the peri-urban and suburban locations are consistent with previously established annual survival in Ohio ($50.5\% \pm 10.3$ SE to $78.3\% \pm 7.6$ SE) and in Washington D.C. (between 40 to 70%) in the United States

(Shustack 2008; Evans et al. 2015). However, the low survival rates that we are seeing in the urban location appear notably lower than rates previously reported for urban habitats, though direct comparisons are complicated by differences in study design, season and the definition given to apparent survival across studies. Cardinals' annual apparent survival was even greater as the amount of impervious surfaces (used as indicator of urbanization) increases from low to moderate in Washington D.C. (Evans et al. 2015). The lower survival in the urban environment is likely attributable to a higher concentration of anthropogenic threats; it was the only site with confirmed window collisions and indeed 67% of confirmed mortality were window collisions. Urban areas present multiple mortality risks for birds, including window collisions, strikes by vehicle and predation by domestic cats (Loss et al. 2014; Ibáñez-Álamo et al. 2020; Lopes et al. 2024). These risks also apply in suburban areas, but to a lesser extent. Specifically, our urban site had a far higher density of windows than our other sites, with many multi-level low-rise buildings (whereas none at other sites) which we hypothesise could have contributed to elevated collision risk. An alternative explanation is that the individuals translocated to the urban site survived but moved farther away from their release site and were beyond the area monitored ($> 18 \text{ km}^2$). We cannot rule out this explanation, and both processes—true mortality and undetected emigration—may have contributed to the low apparent survival at our urban site. Given the confirmed window collisions and the higher density of anthropogenic hazards, we tentatively favour a mortality driven interpretation but acknowledge this remains uncertain. A further hypothesis is that cardinals faced higher risks by moving in a larger area in the urban habitat, a matrix comprised of many threats including domestic cat predation, window collisions, and vehicle strikes in comparison to the suburban and peri-urban areas.

Small scale of association to habitat regardless of urbanization level

Cardinals were associated with habitat features at different spatial scales across our study sites, with the strongest model support occurring at 50 meters for the peri-urban site, and 25 meters for both the suburban and urban sites. While this difference is modest and should be interpreted cautiously, the overall pattern is consistent with cardinals responding to relatively fine-grained habitat features across all sites, regardless of urbanization level. Because telemetry locations and environmental layers contain some level of spatial uncertainty, and because habitat variables measured at adjacent scales are correlated to some level, the estimated scale for the strongest association should be interpreted as an approximation rather than a precise measure of the scale at which cardinals select habitat features. In particular, the spatial error associated with telemetry locations (10-15 m) approaches the difference between our finest scales (25-50 m), further limiting our ability to distinguish between habitat association at these adjacent scales. This aligns with broader research suggesting that the magnitude and direction of species' responses to urbanization reflects landscape-scale differences within cities (Fidino et al. 2021). Moreover, recent work on migratory birds passing through urban areas via stopover sites highlighted the value of multiscale habitat assessment since different processes can come into play at different spatial resolution (Guo et al. 2023). Although our study design did not test this directly, the slightly smaller scale of the strongest model supported for the suburban and urban sites is consistent with the idea that habitat configuration may influence habitat association in more urbanized areas. More broadly, our data suggest that habitat associations were detected at a relatively fine spatial scale across the urban gradient, which seems to be consistent with previous work indicating that small habitat patches could contribute and support bird communities in urban areas (Dale 2018). Nineteen years of banding data at the peri-urban site demonstrated that

the proportion of unbanded to banded birds was similar over most of the year, implying a mostly resident population, yet there was a distinct peak of new birds in late fall (Fig. S3), suggesting a local, 'very short-distance' migration (including post-fledging dispersal) for cardinals. Our findings highlight the potential value of considering multiple spatial scales when evaluating habitat use in urban areas, and are consistent with previous studies suggesting that fine-scale habitat characteristics may be relevant to resident species in urbanized habitats (Litteral and Shochat 2017; Guo et al. 2023). In our case, bird feeders and vegetation structure had an important impact from the smallest scales (third order habitat selection) to the entire range (first order habitat selection).

Variation of habitat associations across an urbanization gradient

Cardinals were associated with different environmental and anthropogenic habitat variables across an urban gradient, indicating that habitat associations are different among the peri-urban, suburban and urban sites. The one consistent association across study sites was the relationship between cardinals' presence and a shorter distance to the closest bird feeder—which is indeed apparent even at the scale of the entire range (Fig. 2; Robb et al. 2008). This is unsurprising as cardinals are one of the most common species at bird feeders within its range, especially during autumn and winter (Horn et al. 2014). We also found a significant relationship with another anthropogenic variable, wherein cardinals were more likely to be found in areas with higher road cover (which may reflect higher residential development) in both the suburban and urban sites. This pattern is consistent with previous work showing a positive association between cardinals and urbanized landscapes during the breeding season (Evans et al. 2015). Comparatively, cardinal associations with natural cover differed across our urbanization gradient. At the urban site, cardinals avoided the areas of highly contrasted vegetation structures, while at the peri-

urban site they were more likely to occupy an area with an elevated standard deviation. These findings suggest that vegetation structure may be associated with winter habitat use, which is consistent with a previous study in Delaware and Maryland identifying that vegetation structure is a particularly important variable in urban areas, specifically for cardinals during the breeding season (Duren et al. 2017). Post-fledging cardinals in urban areas strongly select for habitat with a complex understory vegetation (Ausprey and Rodewald 2011). We did not find evidence of the same association in the peri-urban site. The reason for this discrepancy remains unclear, although variation in habitat availability or habitat heterogeneity among sites may be contributing factors. The importance of vegetation architecture and composition in birds' habitat selection is most likely scale-dependent (Deppe and Rotenberry 2008). Our scale of effect analysis suggests that cardinals selected their wintering habitats at a larger scale at the peri-urban location, compared to the urban site. It is possible that scale-dependent differences in vegetation cues contribute to this pattern, although we caution that the difference between the scales (25 m vs. 50 m) is modest and that this interpretation remains speculative without a direct test of the vegetation composition effect.

Implications

We highlight the importance of fine-scale habitat features and resource availability in shaping winter ecology of a resident songbird along an urban gradient. The context-dependent responses we observed underscore the importance of considering local habitat conditions and resources availability when managing UGS for resident bird species. In particular, the reliance on bird feeders and avoidance of large public UGS suggests that private UGS may play a disproportionately important role in supporting certain overwintering birds. This highlights the often-overlooked importance of private UGS in conservation, even though they are a dominant

and highly diverse part of nature in cities (Goddard et al. 2010; Hutt-Taylor and Ziter 2022). While our study species is not one of conservation concern, it is culturally recognized and may serve as an important species to connect people to nature in urban areas. As urban development intensifies, integrating biodiversity-friendly practices such as preserving vertically complex vegetation and enhancing connectivity between small habitat patches may help sustain populations of resident birds that rely on UGS during the winter. Together, our results highlight how habitat association by a resident bird species is shaped by both landscape-level context and fine-scale resource availability. This multi-scale perspective demonstrates the importance of considering hierarchical habitat use in urban ecology and conservation. More broadly, our findings join a growing body of work emphasizing the ecological importance of urban landscape's elements and highlight the need for inclusive UGS planning that includes both public and private green infrastructures.

Literature Cited

- Altwegg R, Jenkins A, Abadi F (2014) Nestboxes and immigration drive the growth of an urban Peregrine Falcon *Falco peregrinus* population. *Ibis* 156:107–115. <https://doi.org/10.1111/ibi.12125>
- Ausprey IJ, Rodewald AD (2011) Postfledging Survivorship and Habitat Selection Across A Rural-To-Urban Landscape Gradient. *The Auk* 128:293–302. <https://doi.org/10.1525/auk.2011.10158>
- Barron DG, Weatherhead PJ, Brawn JD (2013) A Test of Radio-Transmitter Effects on Parental Investment and Productivity in the Northern Cardinal. *The Condor* 115:669–676. <https://doi.org/10.1525/cond.2013.120149>
- Beddall B (1963) Range Expansion of the Cardinal and Other Birds in the Northeastern States. *Wilson Bull* 75:140–158
- Bressler SA, Diamant ES, Tingley MW, Yeh PJ (2020) Nests in the cities: adaptive and non-adaptive phenotypic plasticity and convergence in an urban bird. *Proc R Soc B Biol Sci* 287:20202122. <https://doi.org/10.1098/rspb.2020.2122>
- Burnham KP, Anderson DR (2002) *Model Selection and Multimodel Inference*, 2nd edn.
- Calenge C (2023) *Home Range Estimation in R: the adehabitatHR Package*
- Calenge C (2024) *adehabitatHR: Home Range Estimation*
- Callaghan CT, Cornwell WK, Poore AGB, et al (2021) Urban tolerance of birds changes throughout the full annual cycle. *J Biogeogr* 48:1503–1517. <https://doi.org/10.1111/jbi.14093>
- Dale S (2018) Urban bird community composition influenced by size of urban green spaces, presence of native forest, and urbanization. *Urban Ecosyst* 21:1–14. <https://doi.org/10.1007/s11252-017-0706-x>
- DeCandido R, Allen D (2006) Nocturnal hunting by Peregrine Falcons at the Empire State building, New York City. *Wilson J Ornithol* 118:53–58. [https://doi.org/10.1676/1559-4491\(2006\)118%5B0053:NHBPFA%5D2.0.CO;2](https://doi.org/10.1676/1559-4491(2006)118%5B0053:NHBPFA%5D2.0.CO;2)
- Deppe JL, Rotenberry JT (2008) Scale-dependent habitat use by fall migratory birds: vegetation structure, floristics, and geography. *Ecol Monogr* 78:461–487. <https://doi.org/10.1890/07-0163.1>
- Dinh KV, Albin D, Orr JA, et al (2023) Winter is coming: Interactions of multiple stressors in winter and implications for the natural world. *Glob Change Biol* 29:6834–6845. <https://doi.org/10.1111/gcb.16956>

599 Duren AM, Williams CK, D'amico V (2017) Microhabitat Factors Associated with Occupancy of
600 Songbirds in Suburban Forest Fragments in the Eastern United States. *Am Midl Nat*
601 178:189–202. <https://doi.org/10.1674/0003-0031-178.2.189>

602 Elmqvist T, Fragkias M, Goodness J, et al (2013) Urbanization, biodiversity and ecosystem
603 services: challenges and opportunities: a global assessment. Springer Nature

604 Environment and Climate Change Canada Historical Climate Data - [Montréal/Pierre Elliott
605 Trudeau International Airport]

606 Evans BS, Ryder TB, Reitsma R, et al (2015) Characterizing avian survival along a rural-to-
607 urban land use gradient. *Ecology* 96:1631–1640. <https://doi.org/10.1890/14-0171.1>

608 Fidino M, Gallo T, Lehrer EW, et al (2021) Landscape-scale differences among cities alter
609 common species' responses to urbanization. *Ecol Appl* 31:e02253.
610 <https://doi.org/10.1002/eap.2253>

611 Fortin D, Beyer HL, Boyce MS, et al (2005) Wolves influence elk movements: behavior shapes a
612 trophic cascade in Yellowstone National Park. *Ecology* 86:1320–1330.
613 <https://doi.org/10.1890/04-0953>

614 Fortin D, Fortin M-E, Beyer HL, et al (2009) Group-size-mediated habitat selection and group
615 fusion–fission dynamics of bison under predation risk. *Ecology* 90:2480–2490.
616 <https://doi.org/10.1890/08-0345.1>

617 Frei B, Morales A, Cousineau A-M, Beauchamp A (2024) Montreal Northern Cardinal Winter
618 Movements (Project 882)

619 Fristoe TS (2015) Energy use by migrants and residents in North American breeding bird
620 communities. *Glob Ecol Biogeogr* 24:406–415. <https://doi.org/10.1111/geb.12262>

621 Goddard MA, Dougill AJ, Benton TG (2010) Scaling up from gardens: biodiversity conservation
622 in urban environments. *Trends Ecol Evol* 25:90–98.
623 <https://doi.org/10.1016/j.tree.2009.07.016>

624 Government and Municipalities of Quebec (2016) Lidar - Digital models (terrain, canopy, slope,
625 level curve) - Open Government Portal. [https://open.canada.ca/data/en/dataset/5e5d2750-](https://open.canada.ca/data/en/dataset/5e5d2750-d129-4952-b9fc-f824c5e480b4)
626 [d129-4952-b9fc-f824c5e480b4](https://open.canada.ca/data/en/dataset/5e5d2750-d129-4952-b9fc-f824c5e480b4). Accessed 17 Nov 2025

627 Government of Canada (2004) Extreme cold. In: Extreme Cold.
628 [https://www.canada.ca/en/health-canada/services/healthy-living/your-](https://www.canada.ca/en/health-canada/services/healthy-living/your-health/environment/extreme-cold.html)
629 [health/environment/extreme-cold.html](https://www.canada.ca/en/health-canada/services/healthy-living/your-health/environment/extreme-cold.html). Accessed 21 June 2026

630 Guo F, Buler JJ, Smolinsky JA, Wilcove DS (2023) Autumn stopover hotspots and multiscale
631 habitat associations of migratory landbirds in the eastern United States. *Proc Natl Acad*
632 *Sci* 120:e2203511120. <https://doi.org/10.1073/pnas.2203511120>

633 Hahs AK, Fournier B, Aronson MFJ, et al (2023) Urbanisation generates multiple trait
634 syndromes for terrestrial animal taxa worldwide. *Nat Commun* 14:4751.
635 <https://doi.org/10.1038/s41467-023-39746-1>

636 Havlíček J, Riegert J, Fuchs R (2022) A comparison of foraging-range sizes, flight distances and
637 foraging habitat preferences in urban and rural House Sparrow (*Passer domesticus*)
638 populations. *Ibis* 164:1227–1242. <https://doi.org/10.1111/ibi.13072>

639 HealthyDesign.City (2021) HealthyPlan.City

640 Horn DJ, Johansen SM, Wilcoxon TE (2014) Seed and feeder use by birds in the United States
641 and Canada. *Wildl Soc Bull* 38:18–25. <https://doi.org/10.1002/wsb.365>

642 Hosmer DW, Lemeshow S (1980) Goodness of fit tests for the multiple logistic regression
643 model. *Commun Stat - Theory Methods* 9:1043–1069.
644 <https://doi.org/10.1080/03610928008827941>

645 Humbird SK, Neudorf DLH (2008) The effects of food supplementation on extraterritorial
646 behavior in female Northern Cardinals. *The Condor* 110:392–395.
647 <https://doi.org/10.1525/cond.2008.8427>

648 Hutt-Taylor K, Ziter CD (2022) Private trees contribute uniquely to urban forest diversity,
649 structure and service-based traits. *Urban For Urban Green* 78:127760.
650 <https://doi.org/10.1016/j.ufug.2022.127760>

651 Ibáñez-Álamo JD, Morelli F, Benedetti Y, et al (2020) Biodiversity within the city: Effects of
652 land sharing and land sparing urban development on avian diversity. *Sci Total Environ*
653 707:135477. <https://doi.org/10.1016/j.scitotenv.2019.135477>

654 Johnson DH (1980) The Comparison of Usage and Availability Measurements for Evaluating
655 Resource Preference. *Ecology* 61:65–71. <https://doi.org/10.2307/1937156>

656 Jokimäki J, Kaisanlahti-Jokimäki M-L, Suhonen J (2022) Long-Term Winter Population Trends
657 of Corvids in Relation to Urbanization and Climate at Northern Latitudes. *Animals*
658 12:1820. <https://doi.org/10.3390/ani12141820>

659 Kauffman MJ, Frick WF, Linthicum J (2003) Estimation of habitat-specific demography and
660 population growth for Peregrine Falcons in California. *Ecol Appl* 13:1802–1816.
661 <https://doi.org/10.1890/01-5324>

662 Latimer CE, Zuckerberg B (2021) Habitat loss and thermal tolerances influence the sensitivity of
663 resident bird populations to winter weather at regional scales. *J Anim Ecol* 90:317–329.
664 <https://doi.org/10.1111/1365-2656.13332>

665 Li G, Fang C, Li Y, et al (2022) Global impacts of future urban expansion on terrestrial
666 vertebrate diversity. *Nat Commun* 13:1628. <https://doi.org/10.1038/s41467-022-29324-2>

667 Litteral J, Shochat E (2017) The Role of Landscape-Scale Factors in Shaping Urban Bird
668 Communities. In: Murgui E, Hedblom M (eds) Ecology and Conservation of Birds in
669 Urban Environments. Springer International Publishing, Cham, pp 135–159

670 Lopes AC, Rocha GOC, de Oliveira Passos MF, et al (2024) The Influence of Building
671 Surroundings and Glass Cover in Bird Collisions. *Birds* 5:703–711.
672 <https://doi.org/10.3390/birds5040048>

673 Loss SR, Will T, Marra PP (2014) Estimation of bird-vehicle collision mortality on U.S. roads. *J*
674 *Wildl Manag* 78:763–771. <https://doi.org/10.1002/jwmg.721>

675 Lowry H, Lill A, Wong BBM (2013) Behavioural responses of wildlife to urban environments.
676 *Biol Rev* 88:537–549. <https://doi.org/10.1111/brv.12012>

677 Manly BF, McDonald LL, Thomas DL, et al (2002) Resource selection by animals: statistical
678 design and analysis for field studies. Springer

679 Martin AE, Fahrig L (2012) Measuring and selecting scales of effect for landscape predictors in
680 species–habitat models. *Ecol Appl* 22:2277–2292. <https://doi.org/10.1890/11-2224.1>

681 Murray MH, Sánchez CA, Becker DJ, et al (2019) City sicker? A meta-analysis of wildlife health
682 and urbanization. *Front Ecol Environ* 17:575–583. <https://doi.org/10.1002/fee.2126>

683 National Audubon Society (2020) The Christmas Bird Count Historical Results [Online].
684 <https://www.audubon.org/community-science/christmas-bird-count>. Accessed 16 Apr
685 2025

686 O'Donnell K, delBarco-Trillo J (2020) Changes in the home range sizes of terrestrial vertebrates
687 in response to urban disturbance: a meta-analysis. *J Urban Ecol* 6:juaa014.
688 <https://doi.org/10.1093/jue/juaa014>

689 Poirier V, Elliott KH, Frei B (2024) Peri-urban forest margins are important stopover sites for
690 moult-migrating songbirds. *J Nat Conserv* 77:126539.
691 <https://doi.org/10.1016/j.jnc.2023.126539>

692 R Core Team (2024) R: A Language and Environment for Statistical Computing

693 Ritchison G, Omer MK (1990) Winter Behavior of Northern Cardinals. *Trans Ky Acad Sci*
694 51:145–153

695 Robb GN, McDonald RA, Chamberlain DE, Bearhop S (2008) Food for thought: supplementary
696 feeding as a driver of ecological change in avian populations. *Front Ecol Environ* 6:476–
697 484. <https://doi.org/10.1890/060152>

698 Robert M, Hachey M-H, Lepage D, Couturier A (2019) Second Atlas of the Breeding Birds of
699 Southern Québec

700 Šálek M, Grill S, Riegert J (2020) Nest-site selection of an avian urban exploiter, the Eurasian
 701 magpie *Pica pica*, across the urban-rural gradient. *J Vertebr Biol* 70:.
 702 <https://doi.org/10.25225/jvb.20086>

703 Seress G, Liker A (2015) Habitat urbanization and its effects on birds. *Acta Zool Acad Sci Hung*
 704 61:373–408

705 Shustack DP (2008) Reproductive Timing of Passerines in Urbanizing Landscapes. The Ohio
 706 State University

707 Simberloff RA, Phillips JN, Derryberry GE, et al (2023) Communication distance predicts
 708 territory size: implications for an urban songbird. *Anim Behav* 203:89–99.
 709 <https://doi.org/10.1016/j.anbehav.2023.05.016>

710 Simkin RD, Seto KC, McDonald RI, Jetz W (2022) Biodiversity impacts and conservation
 711 implications of urban land expansion projected to 2050. *Proc Natl Acad Sci*
 712 119:e2117297119. <https://doi.org/doi:10.1073/pnas.2117297119>

713 Smith BT, Escalante P, Hernández Baños BE, et al (2011) The role of historical and
 714 contemporary processes on phylogeographic structure and genetic diversity in the
 715 Northern Cardinal, *Cardinalis cardinalis*. *BMC Evol Biol* 11:136.
 716 <https://doi.org/10.1186/1471-2148-11-136>

717 Turcotte Y, Desrochers A (2005) Landscape-dependent distribution of northern forest birds in
 718 winter. *Ecography* 28:129–140

719 Vangestel C, Braeckman BP, Matheve H, Lens L (2010) Constraints on home range behaviour
 720 affect nutritional condition in urban house sparrows (*Passer domesticus*): ranging
 721 behaviour in urban House Sparrows. *Biol J Linn Soc* 101:41–50.
 722 <https://doi.org/10.1111/j.1095-8312.2010.01493.x>

723 Vaz S, Manes S, Khattar G, et al (2023) Global meta-analysis of urbanization stressors on insect
 724 abundance, richness, and traits. *Sci Total Environ* 903:165967.
 725 <https://doi.org/10.1016/j.scitotenv.2023.165967>

726 Ville de Montréal (2018) Chaussée agrégée et intersection (Base de données des actifs de voirie)

727 von Bergmann J, Shkolnik D, Jacobs A (2021) censensus: R package to access, retrieve, and
 728 work with Canadian Census data and geography

729 Wang H, He Q, Liu X, et al (2012) Global urbanization research from 1991 to 2009: A
 730 systematic research review. *Landsc Urban Plan* 104:299–309

731 WeatherSpark Climate and Average Weather Year-Round in Montreal. In: *Clim. Aver. Weather*
 732 *Year-Round Montr.* [https://weatherspark.com/y/25077/Average-Weather-in-](https://weatherspark.com/y/25077/Average-Weather-in-Montr%C3%A9al-Quebec-Canada-Year-Round)
 733 [Montr%C3%A9al-Quebec-Canada-Year-Round](https://weatherspark.com/y/25077/Average-Weather-in-Montr%C3%A9al-Quebec-Canada-Year-Round). Accessed 21 June 2026

Worton BJ (1989) Kernel Methods for Estimating the Utilization Distribution in Home-Range Studies. *Ecology* 70:164–168. <https://doi.org/10.2307/1938423>

Supplemental material

Cardinals tagging procedure

We mist netted cardinals (males and females) at the McGill Bird Observatory (MBO) during October and November of the year 2022, 2023 and 2024. The birds were intercepted using 30 mm Japanese Nylon nets and placed in small cotton bags to reduce stress before handling. As part of the migration monitoring being done at the MBO, we banded all cardinals with a numbered aluminum band issued by the Canadian Bird Banding Office. The individuals were sexed, aged, weighed, and randomly assigned to one of the six possible release sites consisting of translocation sites and control sites at the capture location. In addition to the aluminum band, we fitted all cardinals with a set of heat-sealed coloured plastic bands, each with a unique combination of colours. This allowed visual identification of the individuals without having to resort to a radio tag receiver. Each bird was equipped with a Lotek NanoTag (NTQB-2) weighing 0.8 grams, using soft elastics in a figure 8 harness configuration (the total weight of the tag, harness, and glue was less than 3% of cardinals' body weight). All the radio-tags were registered as part of the Motus Wildlife Tracking System (Frei et al. 2024).

Motus tower locations

During the first winter, only 2 automated receiver towers from the Motus network were deployed. The first tower is located at the peri-urban site (45°25'50.4"N 73°56'18.6"W), while the second tower is located on top of a faculty building at McGill University's Macdonald campus (45°24'28.3"N 73°56'20.3"W). For the second winter, those two towers were also used for continuous data location, with the addition of a third temporary receiver tower that we

installed at the suburban site (45°24'39.3"N 73°55'48.3"W). Finally, during the third winter, a fourth, permanent Motus tower was installed on top of a building at Concordia University's Loyola campus location at the urban site (45°27'32.6"N 73°38'30.2"W).

Detailed use of gain and signal strength when tracking

The gain is a value between 0 and 100, representing the amplification of the received signal. By increasing it, the tracker increases how sensitive the receiver is to incoming radio signals. Higher gain values increase the ability to detect weaker signals but are also subject to background noise and interference. Using a lower gain reduces noise but may result in missing weak signals. In urbanized locations, the starting gain will rarely surpass 70-80 as the receiver picks up too much noise and interference. The signal is a value ranging from 0 to 200 for the SRX800 and 0 to 250 for the SRX1200. It represents the strength of the detected transmission from a radio tag, displayed in decibels (dB) on the receiver. A stronger signal indicates that the bird is closer; trackers use this information to estimate distance and direction. Trackers begin their tracking sessions with a gain of 80-70 to increase their chances of detecting nearby individuals. Once the receiver picks up a cardinal's tag, the tracker moves toward the direction with the strongest signal. Once a threshold of 150 dB (SRX800) or 200 dB (SRX1200) is attained, the trackers then lower the gain by 10 and repeat this process until they reach the tagged individual. The trackers only record a GPS point if the signal strength is 150 or 200 dB, with a gain of 40 or less. While visual confirmation was difficult to achieve due to the nature of the release habitats (high density of private property), prior testing with radio tags showed that we could assume the bird to be within 10-15 meters when the gain was ≤ 30 and the signal strength ≥ 150 dB for the SRX800 receivers (or 200 dB for the SRX1200). The handheld receivers were either Lotek's SRX800 or

781 SRX1200. They were operated by setting the frequency to 166.380 MHz (Motus tags' frequency
782 for the Western hemisphere) and using both the gain and the signal strength to get as close as
783 possible to the tagged individuals. We were usually able to detect birds that were within 200
784 meters of our location, and on some rare occasions were even able to pick up signals of birds
785 located as far as 800 meters away. We did encounter a significantly higher level of static noise
786 when tracking at the urban location, yet it was not enough to prevent us from finding birds within
787 the detection range. Another factor that could have complicated tracking efforts was the
788 occasional aggregation of multiple birds, which can make it difficult to obtain precise location
789 data. In most cases, no more than two birds were observed together. When larger groups were
790 detected, the majority typically dispersed before we could approach close enough to record a
791 GPS location. Overall, however, such aggregations were infrequent and are therefore unlikely to
792 have significantly affected the accuracy of our tracking data.
793

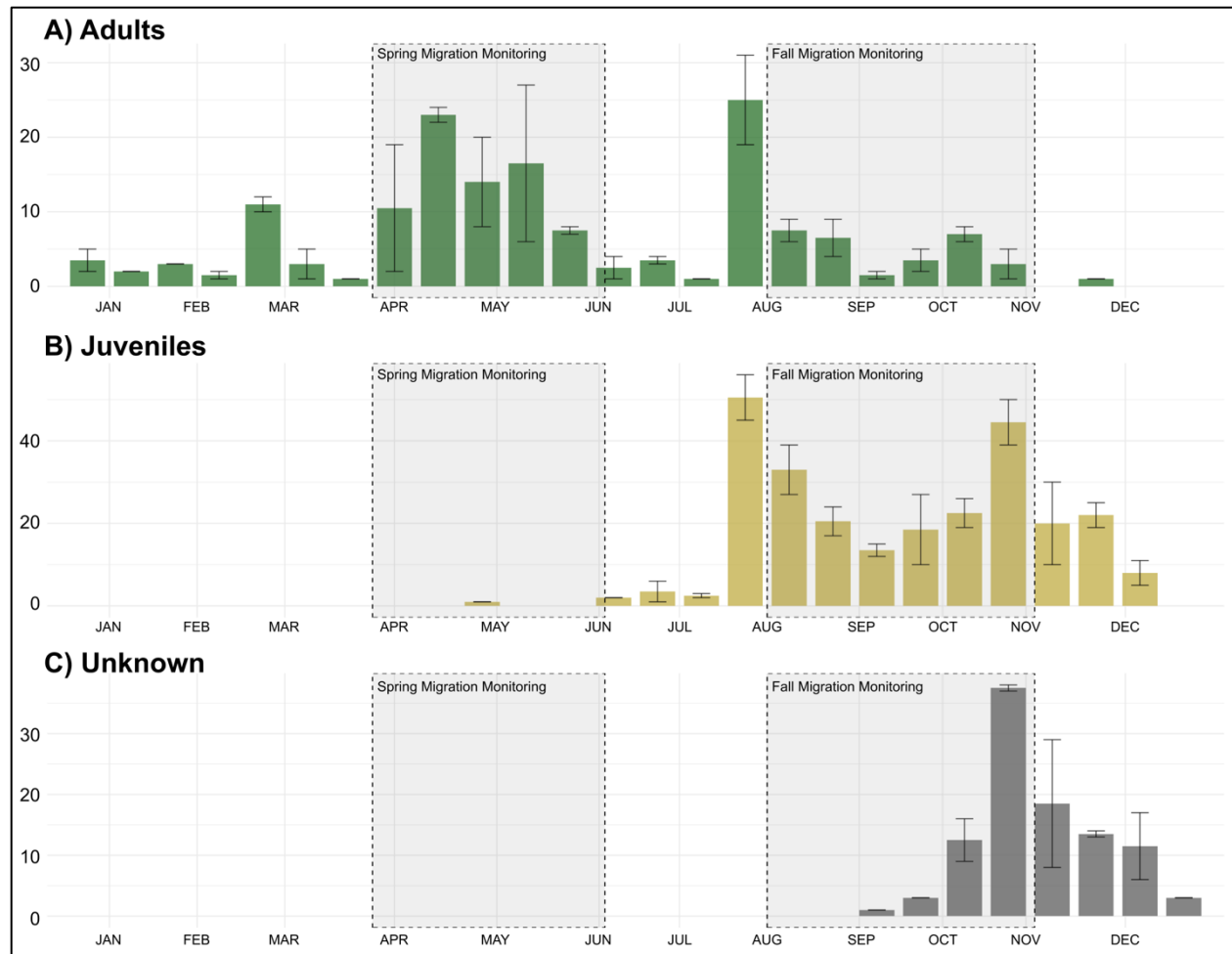


Figure S1. Mean number of cardinals captured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada separated by age where A) shows adult birds, B) shows juvenile birds and C) shows unknown individuals.

Table S1. Model ranking using AICc of generalized linear mixed models (GLM) with a comparing scale-of-effect of landscape variables for resighting locations (n= 57) of cardinals tracked across three sites in Montreal, Quebec, Canada.

Study site	Model	Variables	LL	Δ AICc	Wt
Peri-urban	25 meters	Mean vegetation height + sd	-722.48	17.02	0.00
	50 meters	vegetation height + mean	-713.97	0.00	1.00
	100 meters	population density + road	-735.69	53.25	0.00
	200 meters	length + distance to nearest feeder	-740.60	46.96	0.00
Suburban	25 meters	Mean vegetation height + sd	-296.05	0.00	1.00
	50 meters	vegetation height + mean	-338.31	84.52	0.00
	100 meters	population density + road	-344.25	96.39	0.00
	200 meters	length + distance to nearest feeder	-349.92	107.73	0.00
Urban	25 meters	Mean vegetation height + sd	-211.09	0.00	1.00
	50 meters	vegetation height + mean	-225.64	29.11	0.00
	100 meters	population density + road	-224.02	25.85	0.00
	200 meters	length + distance to nearest feeder	-228.79	35.40	0.00

Table S2. AIC values used to determine the best fitting model to explain cardinal's habitat associations.

Study site	Model	Variables	AICc
Peri-urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	1443.696
	Vegetation height	Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	1587.124
	Vegetation sd	Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	1596.077
	Human	Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	1588.262
	Null	(1 Winter) + (1 ID) + (1 Site)	1616.101
Suburban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	631.7501
	Vegetation height	Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	728.0640
	Vegetation sd	Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	724.0187
	Human	Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	632.1451
	Null	(1 Winter) + (1 ID) + (1 Site)	726.1005
Urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1 Winter) + (1 ID) + (1 Site)	444.7050
	Vegetation height	Mean vegetation height+ (1 Winter) + (1 ID) + (1 Site)	486.6402
	Vegetation sd	Sd vegetation height+ (1 Winter) + (1 ID) + (1 Site)	485.2496
	Human	Mean population density + road length + distance to nearest feeder+ (1 Winter) + (1 ID) + (1 Site)	447.8691
	Null	(1 Winter) + (1 ID) + (1 Site)	484.8853